

WS N°3: Phase Diagrams (Part II)

- Is it possible to have a copper–silver alloy of composition 50 wt% Ag–50 wt% Cu that, at equilibrium, consists of α and β phases having mass fractions $W_\alpha = 0.60$ and $W_\beta = 0.40$? If so, what will be the approximate temperature of the alloy ? If such an alloy is not possible, explain why.

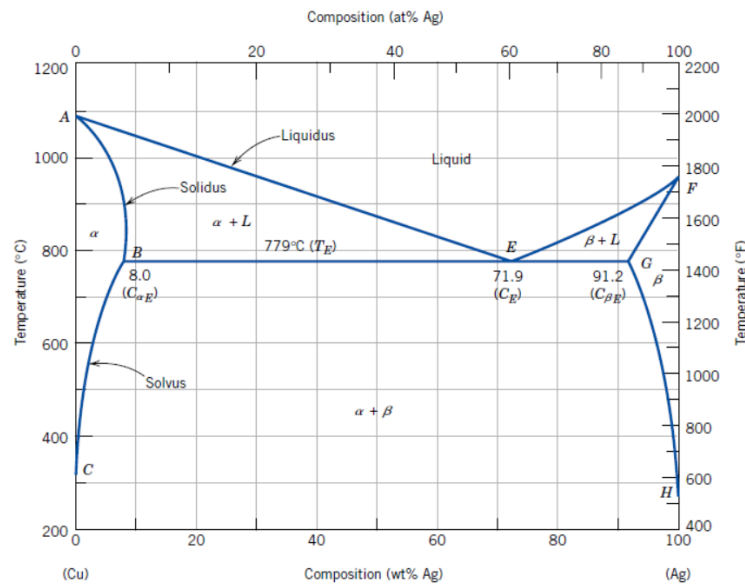


Figure 9.7 The copper–silver phase diagram. [Adapted from *Binary Alloy Phase Diagrams*, 2nd edition, Vol. 1, T. B. Massalski (Editor-in-Chief), 1990. Reprinted by permission of ASM International, Materials Park, OH.]

- A 45 wt% Pb–55 wt% Mg alloy is rapidly quenched to room temperature from an elevated temperature in such a way that the high temperature microstructure is preserved. This microstructure is found to consist of the α phase and Mg_2Pb , having respective mass fractions of 0.65 and 0.35. Determine the approximate temperature from which the alloy was quenched.

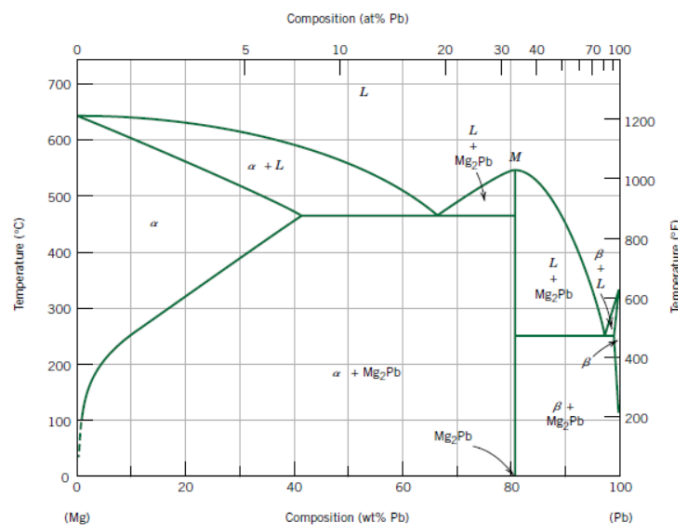


Figure 9.20 The magnesium–lead phase diagram. [Adapted from *Phase Diagrams of Binary Magnesium Alloys*, A. A. Nayeb-Hashemi and J. B. Clark (Editors), 1988. Reprinted by permission of ASM International, Materials Park, OH.]

3. For a copper–silver alloy of composition 25 wt% Ag–75 wt% Cu and at 775°C (1425°F), do the following:
- Determine the mass fractions of α and β phases.
 - Determine the mass fractions of primary α and eutectic microconstituents.
 - Determine the mass fraction of eutectic α .

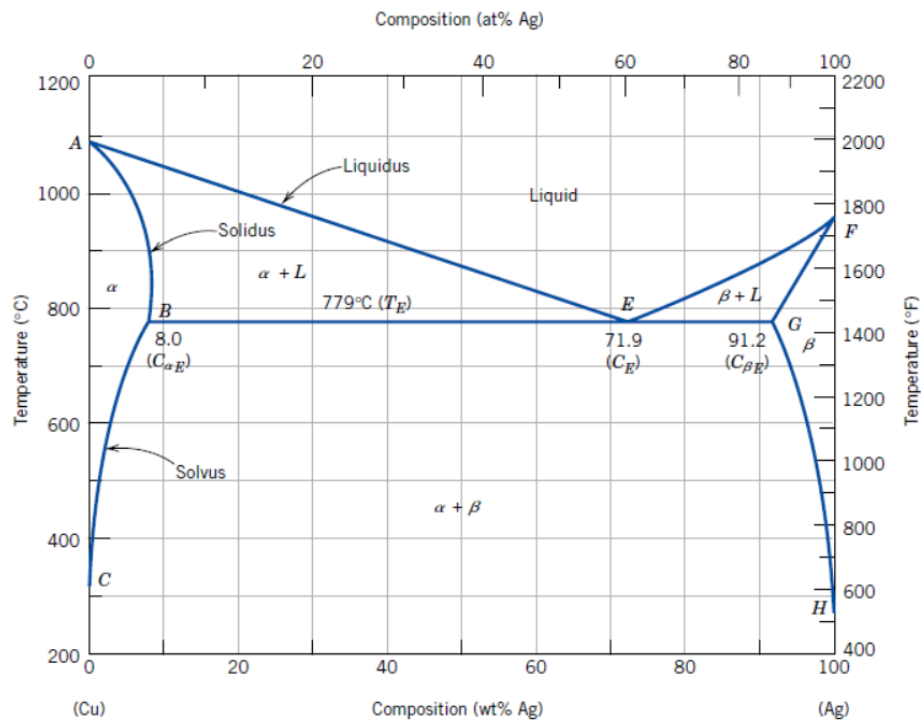
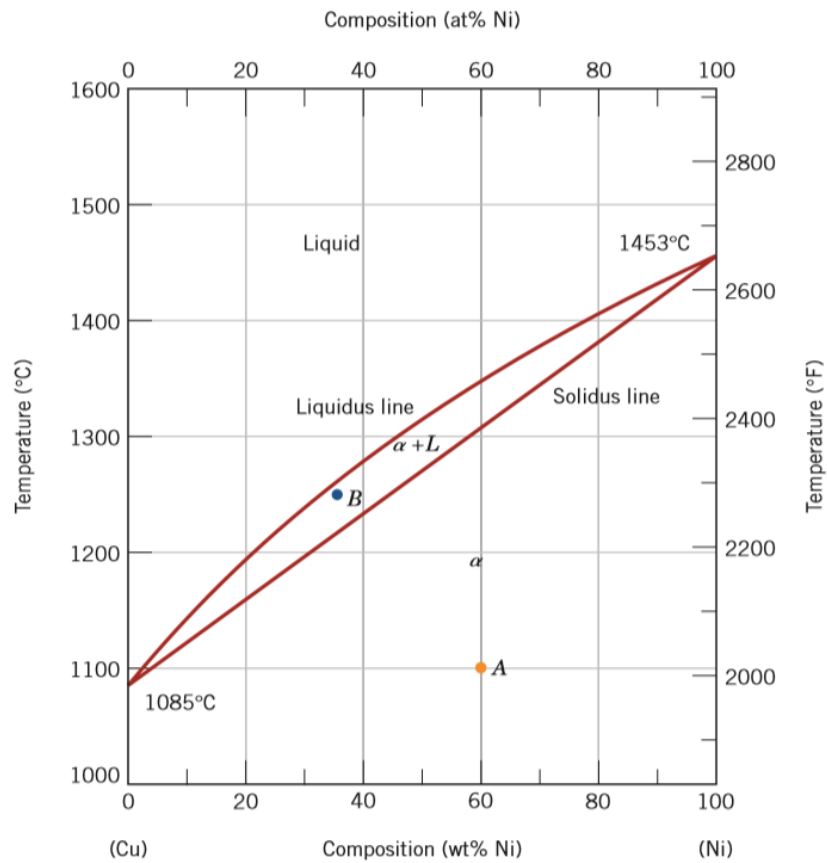


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4. A copper–nickel alloy of composition 70 wt% Ni–30 wt% Cu is slowly heated from a temperature of 1300°C (2370°F).
- At what temperature does the first liquid phase form?
 - What is the composition of this liquid phase?
 - At what temperature does complete melting of the alloy occur?
 - What is the composition of the last solid remaining prior to complete melting?



5. A 90 wt% Ag-10 wt% Cu alloy is heated to a temperature within the β + liquid phase region. If the composition of the liquid phase is 85 wt% Ag, determine:
- The temperature of the alloy
 - The composition of the β phase
 - The mass fractions of both phases

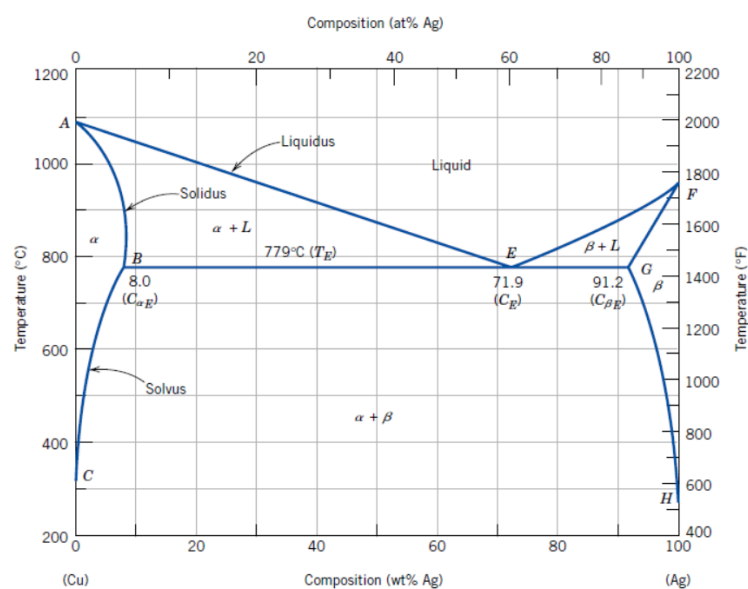


Figure 9.7 The copper-silver phase diagram. [Adapted from *Binary Alloy Phase Diagrams*, 2nd edition, Vol. 1, T. B. Massalski (Editor-in-Chief), 1990. Reprinted by permission of ASM International, Materials Park, OH.]